European Polymer Journal 214 (2024) 113153

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Review

Advantages of animal leather over alternatives and its medical applications

Hsiuying Wang

Institute of Statistics, National Yang Ming Chiao Tung University, Hsinchu 300093, Taiwan

ARTICLE INFO

Keywords:
Animal-based leather
Genuine leather
Medical application
Synthetic leather
Vegan leather

ABSTRACT

The leather industry, grounded in ancient practices, utilizes animal skins composed of collagen protein polymer as materials. This versatile animal-based material is crucial for crafting items like shoes, clothing, and car interiors. While animal-based leather has faced criticisms for environmental challenges in the production process, including wastewater and solid waste generation, as well as concerns about animal ethics, it holds numerous merits compared to synthetic leather and plant-based vegan leather. This paper reviews the advantages of animal-based leather and its potential medical application. The widespread use of animal leather reveals that, despite the passage of time, it remains an important material for humans. The benefits of animal-based leather, in comparison to synthetic leather and plant-based vegan leather, encompass biodegradability, biocompatibility, natural feel and aesthetics, breathability, durability, and longevity. Animal-based leather has been utilized in various medical applications, including pharmaceuticals, sensors, and medical implants.

These advantages could make animal-based leather more suitable for manufacturing medical implants compared to alternatives. Therefore, this paper suggests further development of medical equipment and materials based on animal-based leather to promote human health.

1. Introduction

The production of leather is one of the oldest industries, involving the effective use of animal skins. Comprised of collagen protein polymer, animal-based leather plays a crucial role in crafting various items such as shoes, car interiors, clothing, and sports equipment. The use of leather dates back to ancient times when humans practiced hunting for food [1]. The skins and hides from deceased animal carcasses were utilized for clothing, footwear, and the construction of crude tents for protection. This practice even finds mention in the origin of human history in the Bible: 'The Lord God made garments of skin for Adam and his wife and clothed them [2]. Throughout history, leather goods have held a significant place in human life. Their historical significance in various cultures and societies underscores their enduring appeal and timeless value as materials for various uses.

In ancient times, people observed that raw animal skins and hides would putrefy when exposed to moisture, and become hard and rough upon drying. The inconvenience of the hardened state for wrapping around the body led to efforts to achieve optimal drying, softening the fur, and making the skin more pliable and convenient. This process is known as tanning. In leather production, the tanning process is often considered a significant contributor to waste generation. As global emphasis on environmental protection grows, wastewater, sludge, and solid waste from tanneries have emerged as the three key challenges hindering the sustainable development of the leather industry [3].

Chrome tanning persists as the predominant technology in the global leather industry, appreciated for its capability to create leather endowed with qualities sought after in high-quality products [4]. Despite its widespread use, chrome tanning is associated with significant environmental impacts. A comparison of the environmental effects of various tanning systems was conducted, including a chrome tanning system (CTS) employing basic chrome sulfate (Cr), a chrome-free metal tanning system (CFMTS) utilizing complexes of highly oxidized starch and Al–Zr (HOS–AZ), and a metal-free tanning system (MFTS) employing oxidized sodium alginate (OSA) [5]. The MFTS demonstrated lower environmental impacts in terms of resource consumption and climate change when compared to both the CFMTS and the CTS. The combination of tanning using vegetable tannins and aluminum sulfate presented a promising eco-friendly approach to leather tanning [6].

Animal-based leather discussed in this paper encompasses a wide range of sources, including skins, hides, and organs. Animal-based leather is commonly referred to as genuine leather when it is primarily produced from the skin of animals such as cows, goats, and sheep. However, the term 'genuine leather' usually does not include leather made from fish skin or animal organs. Leather, being a versatile material, has various applications. Animal-based leather is typically

E-mail address: wang@stat.nycu.edu.tw.

https://doi.org/10.1016/j.eurpolymj.2024.113153
Received 4 April 2024; Received in revised form 15 May 2024; Accepted 17 May 2024
Available online 18 May 2024
0014-3057/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

recognized as a high-quality and costly material, commonly used in luxury products such as bags, clothes, shoes, and sports equipment. In addition to its high quality, animal-based leather offers numerous other benefits, including biodegradability and biocompatibility, distinguishing it from synthetic leather and plant-based vegan alternatives. Its inherent biocompatibility with human tissues makes it suitable for various medical applications such as wound dressings, implants, and scaffolds for tissue engineering.

Additionally, the scraps (waste) generated during animal leather processing can also be utilized as raw materials for crafting enhanced consumer goods, including insoles, upper parts for sandals, purses, small handbags, mouse pads, and various interior decorations [7]. Furthermore, the scraps can be employed in the manufacturing of fertilizers for soil conditioning, given their high nitrogen (N) content. Utilizing these waste materials for soil conditioning or fertilizer production is one of the alternatives to diminish waste disposal [8]. Leather scraps can be hydrolyzed, similar to poultry feathers. The resulting biomass is solvent-extracted, dried, and turned into animal meal, serving as both feed and a protein source [9]. The repurposing of leather scraps in various ways, including upcycling and recycling, can minimize waste and maximize resource utilization in the leather industry.

Recent widespread attention has been directed towards climate warming. Consequently, carbon reduction has emerged as a critical global concern [10]. As the population continues to expand, the demand for synthetic leather becomes a substantial contributor to carbon emissions. Animal-based leather, as well as lab-grown or biofabricated leather, may present eco-friendly alternatives that align with the imperative to reduce carbon footprints because they are biodegradable, naturally decomposing at the end of their life cycle without leaving behind harmful pollutants. Fish leather possesses the advantage of not demanding resources or leaving behind the carbon footprint typically associated with the cultivation of cattle or the production of synthetic leather [11]. Additionally, despite considerable efforts directed toward synthesizing materials that replicate natural structures, their intended performance is rarely comparable to that of animal-based leather. Therefore, the direct integration of natural motifs into functional materials offers straightforward pathways to novel properties, conferring practical advantages across diverse engineering and industrial sectors.

This paper reviews the properties of animal-based leather and its medical applications, especially aiming to emphasize the potential of animal-based leather for medical use. During the COVID-19 pandemic, where shortages in medical materials and food prevail, exploring natural biomaterials becomes crucial [12–15]. Beyond its eco-friendly and versatile qualities, animal-based leather holds the potential to be utilized in medical applications, including the development of medical devices or implants. It presents an innovative solution to address the pressing needs of the healthcare sector. Despite the ongoing debate surrounding alternative materials, the benefits of animal-based leather, including biodegradability, biocompatibility, natural feel and aesthetics, breathability, durability, and longevity, maintain its importance and utility. Promoting the exploration of animal-based leather not only tackles immediate shortages but also positions it as a multifaceted resource contributing to both sustainability and advancements in medical technology during these challenging times.

2. Types of leather

2.1. Animal-based leather

The raw materials for the animal-based leather industry are sourced from hides and skins of various animals, including cattle, camels, sheep, and goats. Additionally, exotic sources like ostriches, rabbits, crocodiles, and fish contribute to the diversity of materials used in the industry [16]. In the past, due to material scarcity, leather made from animal skin or hide was used for various applications, including clothing or the creation of bags for storing wine or water. This indicates that, like wood, animal-based leather has been crucial to human civilization. Now, animal-based leather is utilized in a wide range of consumer products, including fashion accessories, apparel, furniture, automotive interiors, and luxury goods.

Skins of cattle, sheep, and goats are commonly used materials for making leather due to the quality and versatility of their hides. Crafted leathers are valued for their durability, softness, and suitability for various applications. Cattle, sheep, or goats are the main sources of leather production in certain countries [17–19]. In some countries, these hides and skins contribute a significant portion of the value of livestock output and are an important source of foreign exchange earnings [20].

Ostrich leather is an important industry in some areas. The effect of age on the physical skin traits of slaughter ostriches was investigated [21]. A structured study was conducted on the properties of ostrich leather, with a focus on factors such as age, nutrition, slaughter weight, and genetics, and their influence on intrinsic leather traits [22]. Age appeared to positively correlate with increased leather thickness and tensile strength. Heavier slaughter weights were associated with higher slit tear strength and tensile strength. Older ostriches exhibited greater values in slit tear strength and skin thickness [22].

Camel hides, often overlooked in comparison to other hides, possess unique durability and grain patterns similar to bovine and buffalo leather. However, challenges like hide damage and awkward shapes hinder utilization. Modern farming and tanning could enhance camel hide marketing [23]. Sudan, with a substantial livestock population, particularly camels, presents opportunities for the leather sector. Recent studies compared camel hides to conventional cow hides, showing comparable properties for shoe upper leather [24]. The dimensions of primary and secondary follicles in camel skin could potentially be used to predict the tensile strength, tearing strength, and elongation of tanned leather [25]. Leather fatliquor was successfully formulated from camel hump fat through a sulfitation process [26].

Fish skin leather is produced by repurposing waste from the food industry into a durable material for fashion and textiles. Similar in strength to cowhide, it is used for various products, and its unique grain structure has attracted interest among tanners for its high market value [27]. The fish skin was upcycled into leather, with a water-protecting coating developed using soybean oil and fatty acid trimer [28]. This green chemistry approach enhances hydrophobicity in leather and cotton without compromising breathability or flexibility, contributing to sustainable alternatives in textile materials. Skin from a star puffer fish in the Indian Ocean was used to make exotic leathers [29]. The leather, distinct due to calcified spines, displayed superior mechanical properties, suitable for high-grip articles and therapeutic footwear. Salmon leather underwent three finishing treatments—hydrophobic, oleophobic, and electro-conductive coatings, which resulted in enhanced versatility [30].

In addition to skin, other animal organs such as the stomach and bladder can be sources for making leather. Exotic leathers, prized for unique patterns and marks, hold significant market value. The tanning of ovine stomachs into exotic leather using goat oil was explored, suggesting a material suitable for crafting distinctive small leather goods [31]. The viability of converting putrescible goat stomachs into leather for small goods was studied, examining physical and chemical properties [32]. Despite its comparatively lower strength, the material presented itself as novelty leather suitable for the production of accessories. In some regions throughout history, animal bladders have been utilized to fashion wine bags and liquid containers. Traditional wine carriers, for instance, were often fashioned from goat bladders and covered in goat skin.

Fig. 1 provides some animal sources discussed in this paper.

2.2. Synthetic leather

Synthetic leather, also referred to as faux leather or artificial leather, is a crafted alternative to animal-based leather. Synthetic leather

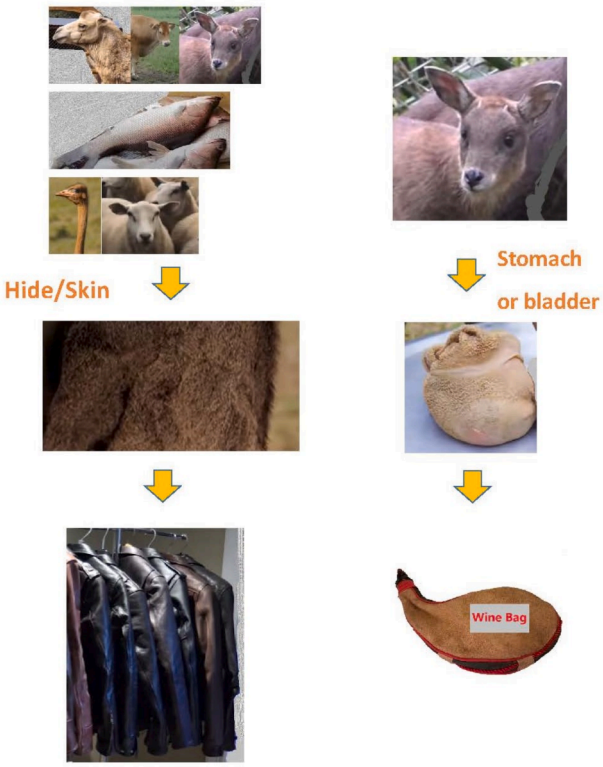

Fig. 1. The animal sources discussed in this paper used in the production of leather products.

typically comprises a foundational textile, such as a knitted, woven, or nonwoven fabric, onto which a polymer is coated in the form of either lacquer or emulsion [33]. Polyurethane (PU) is the most frequently utilized polymer in synthetic leather, while polyvinyl chloride (PVC) and polypropylene are alternative options [34–36]. Fundamentally, the production process of synthetic leather begins with polymerization of materials such as PU or PVC, followed by coating onto a textile fabric substrate. Subsequently, embossing is employed to create texture, with finishing treatments applied thereafter to enhance the properties of the synthetic leather [37].

Synthetic leather is crafted to emulate the appearance, and occasionally the texture, of genuine leather, while typically being more cost-effective. Synthetic polymer materials are readily accessible, contrasting with natural hides. The scarcity and expense of animal-based leather have led to a growing demand for synthetic alternatives that offer similar characteristics such as consistent texture, color, flexibility, and softness [33]. As a consequence, the advancement of synthetic leather has diminished the necessity for animal hides and has been firmly established in various applications [38]. The use of synthetic leather in car interiors is firmly established and has been positively embraced by consumers for its soft and smooth tactility [39].

There are several categories of synthetic leather, such as PU leather, PVC leather, and poromeric leather [40]. Based on the coating type, synthetic leather can be categorized into two groups: PVC artificial leather and PU artificial leather [41]. PU has notable characteristics, including exceptional mechanical strength, effective abrasion resistance, toughness, and flexibility at low temperatures. The advanced technology applied in the coating process of synthetic leather, utilizing PU/graphene composites, possesses remarkable features [42]. Within various applications, PU stands out as an important retanning agent and coating material in leather manufacturing [43].

However, traditional PU is combustible, and its burning process involves intense melting and dripping, leading to the production of harmful fumes and gases [44]. PVC leather is another category of synthetic leather, which, unlike PU leather, is prone to cracking over time and exhibits a plastic-like appearance. It is often employed in the production of low-cost items [45]. The primary component of PVC artificial leather, PVC itself, can lead to soil and environmental pollution through improper disposal [46]. Poromeric leather resembles PU or PVC leather. The sole distinction lies in the fact that poromeric leather is crafted by combining various synthetic fibers to achieve a texture closely resembling genuine leather [40].

Synthetic leather is typically manufactured using petroleum-based materials, which has raised environmental concerns. As consumer awareness of climate change and global warming grows, their preferences have shifted, leading to an increased demand for sustainably manufactured goods. Consequently, this shift has given rise to a new category within the leather industry, commonly known as vegan leather [40].

2.3. Plant-based vegan leather

Vegan leather is a recent innovation in the realm of sustainable fashion, crafted by repurposing leftovers or cultivating biological cells [47]. The term “vegan” inherently refers to food or products that are sourced without the use of animals and do not entail the exploitation or harm of animals [48]. The production of plant-based vegan leather begins with the careful selection of plant materials such as pineapple leaves [49]. These materials undergo processing to extract fibers, which are then formed into sheets. Subsequently, finishing treatments are applied to enhance texture and appearance.

Beyond its departure from animal-derived sources, vegan leather successfully mimics the traits of animal-based leather, offering a practical substitute. The growing awareness of animal welfare over the last few decades of the twentieth century has led to an increased interest among consumers in materials that replicate the look and feel of animal-based leather such as synthetic leather and plant-based vegan leather [50]. While vegan materials address animal ethics, the ongoing concern is the environmental impact of producing and disposing of composite materials, raising environmental pollution issues [51].

Various plants have been used as materials for vegan leather. Fungi-derived leather substitutes, known for their ethical and environmentally responsible production, are gaining popularity due to their aesthetic and functional qualities. Produced through upcycling, they mimic traditional leather without animal or synthetic sources [52,53]. The characteristics of cactus leather have been explored, revealing a potential bio-based alternative that can effectively emulate numerous functions of traditional animal leather [54]. Certain varieties of cactus-based leather are free of PVC and comprise approximately 28 % of bio polyurethane, resulting in a lower environmental impact compared to traditional and synthetic leather [55]. Mango leather was made by separating flesh from the seed, adding binding agents, oven-cooking, finishing, and placing on organic cotton, which showed satisfactory mechanical properties [55]. The cast-tape drying (CTD) method is a fitting and innovative approach for dehydrating fruit pulps, employing moderate temperatures and relatively brief processing times. This results in highly appealing products, such as mango leathers produced in just 18 min using CTD [56]. Three varieties of banana fabric were selected and examined for their physical, morphological, and thermal properties [57]. The results revealed that the majority of the physical properties exhibited by banana-based fabrics were comparable to those of buffalo shrunken grain leather commonly employed in leather goods applications.

The sources of the three types of leather discussed in this section are presented in Fig. 2.

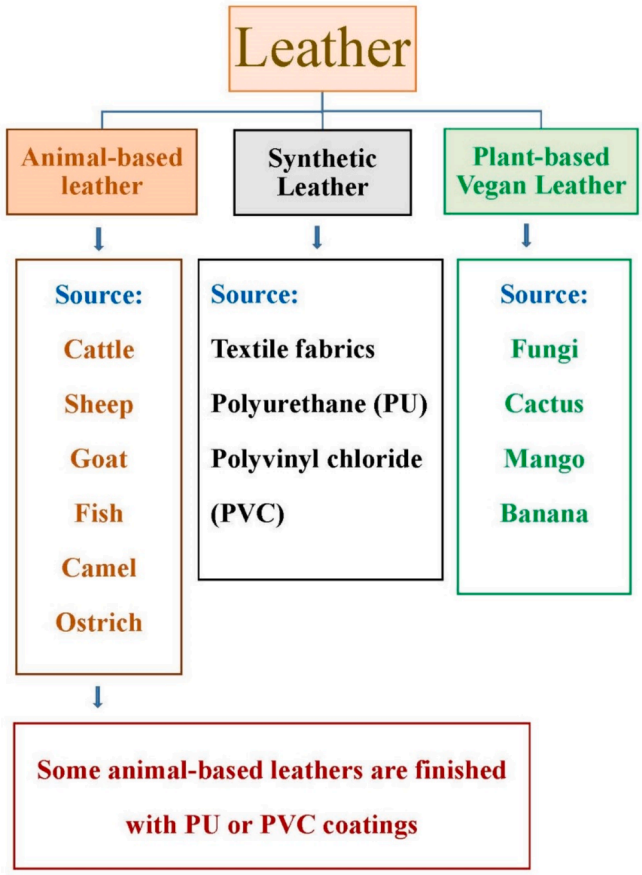

Fig. 2. The sources of the three types of leather discussed in this section.

3. Production and properties of animal-based leather

3.1. Production processes

Animal hides undergo a sequence of chemical processes to be converted into leather, including preparation of raw hides, curing, soaking, liming, deliming and bating, tanning, dyeing, fatliquoring, drying, and finishing [58]. The functions of these processes are listed in Table 1.

Table 1
The processes involve in making animal-based leather and their functions.

Process	Function
Preparation of Raw Hides	Raw animal hides are obtained from slaughterhouses or meat processing plants. These hides undergo initial cleaning processes to remove flesh, hair, and any other impurities.
Curing	The hides are then treated with salt to prevent bacterial growth and preserve them during transportation and storage.
Soaking	Cured hides are soaked in water to rehydrate and remove the salt. This step prepares the hides for further processing.
Liming	Hides are treated with lime to remove hair and epidermis, making the hides suitable for tanning.
Deliming and Bating	Deliming follows to neutralize the pH, and bating is done to soften the hides by using enzymes.
Tanning	Tanning is a crucial step where hides are treated with tanning agents to stabilize the collagen fibers, preventing putrefaction and turning the material into leather.
Dyeing	The leather may undergo a dyeing process to add color according to the desired specifications.
Fatliquoring	This step involves adding fats and oils to the leather to enhance its flexibility and softness.
Drying	The leather is then dried to remove excess moisture, and it may be stretched or tumbled during this phase.
Finishing	The leather goes through finishing processes, including sanding, buffing, and coating, to achieve the desired texture, appearance, and durability.

The main structural component of leather is type I collagen in the form of strong fibrils [59]. The tanning process holds paramount significance as it stabilizes the collagen fibers within the animal skins, thereby averting putrefaction and facilitating the transformation into a resilient leather material [60]. Different tannages are available to meet specific quality preferences for leather. These include mineral tanning options such as chromium, aluminum, iron, and zinc, vegetable tanning with materials like mimosa, acacia, and quebracho, aldehyde tanning involving glutaldehyde and formaldehyde, as well as oil tanning [16]. The utilization of plant materials, including leaves, twigs, fruits, barks, or roots, to prevent the degradation of animal skins and transform them into a more durable and useful material may have commenced towards ancient times, marking the early practices of tanning [61]. The growing emphasis on environmentally friendly practices has elevated the significance of vegetable tannins as key tanning agents in the leather industry. Various types of vegetable tannins have distinct effects. The effects of different types of vegetable tannins including black wattle (Acacia mearnsii), quebracho (Schinopsis lorentzii), chestnut (Castanea sativa), tara (Caesalpinia spinosa), and myrobalan (Terminalia chebula) were evaluated [62].

After the tanning process, the surface treatment of leather includes various techniques and applications aimed at enhancing its properties and appearance. These treatments may involve dyeing, fatliquoring, drying, and finishing. Some animal-based leathers are finished with PU or PVC coatings, or with coatings made from other materials. Compounds such as hydrogels or quantum dots could be developed for use as coating materials [63–65]. In addition, the collagen network of leather creates an environment conducive to microbial growth. Coating agents with antimicrobial properties are therefore recommended [66]. Coatings containing chitosan have been developed specifically to coat leather surfaces and prevent bacterial and fungal attacks [67]. These coatings not only protect the integrity of the leather but also inhibit the growth of microorganisms, thereby enhancing product longevity and safety.

3.2. Properties

The primary attributes that render animal-based leather a valuable product are its strength and flexibility [68]. Strength is an important property of leather, and the strength property of leather is determined by collagen fiber size and the arrangement of collagen fibril bundles. Various animal-based leathers have different properties. Traditional methods for determining the animal source of leather are challenging. An LC-MS method has been suggested to identify interspecies differences in the amino acid sequence of type I collagen, allowing for swift and uncomplicated certification of leather [69]. Synchrotron-based small angle X-ray scattering (SAXS) can be employed to measure the collagen fibril diameter in leather sourced from various animals. A correlation was observed in bovine leather between collagen fibril diameter and tear strength, indicating that stronger leather tended to have thicker fibrils [70]. Nano-biocomposites are complex systems that merge the functions of each component, creating new physico-chemical properties. These materials are crucial in leather processing, enhancing properties and reducing toxic waste, particularly after tanning. A bio-polymeric nanocomposite was synthesized as a retanning agent in leather processing, capable of enhancing the softness and fullness of the leather [71].

In addition, near-infrared spectroscopy (NIRS) models were developed by utilizing physical–mechanical tests to assess the quality of cattle leather [72]. The physical leather quality characteristics of different sheep breeds were investigated [73]. The result revealed no distinctions in leather properties between ewes and rams, while skins from hair-type sheep exhibited stronger properties than those from wool-type sheep. The impact of various roughage-to-concentrate ratios on the mechanical quality of leather in Blackhead Ogaden sheep was examined [74]. The outcome revealed that elevating concentrate levels in the diet enhances leather production within the acceptable range defined by the Ethiopian

leather industry for most test parameters. The attenuated total reflection Fourier-transform infrared (ATR-FTIR) spectroscopy could be a useful tool to examine the chemical structure of sheep leather [75]. Flexural rigidity and thickness, as independent variables, were used to study the drapability of sheep garment leather [76].

Animal-based leather offers benefits similar to biomass materials, including biodegradability and excellent biocompatibility. In contrast, synthetic polymer substrates have lower mechanical properties and wear resistance, posing a potential environmental threat due to their limited biodegradability [3]. Nine vegan leather alternatives were compared with traditional leather [77]. While some alternatives exhibited high values in specific properties. None of these alternatives matches the universal performance of animal-based leather.

Generally, animal-based leather may exhibit superior properties compared to synthetic leather (Table 2). It is known for its natural beauty, rich texture, and longevity. These properties distinguish animal-based leather from other materials, making it a premium material choice for luxury goods. However, a drawback of animal-based leather clothing is the challenge of care, as it can be difficult to clean stains and requires careful maintenance [78].

4. Medical application

4.1. Drug

The waste generated during leather production can be used to extract collagen for medical applications. Skin collagen, characterized as a biomass obtained by eliminating non-collagen constituents from the skin, exhibits exceptional performance in terms of mechanical properties, chemical reactivity, and biocompatibility. These good biological properties of skin collagen such as excellent biocompatibility, low antigenicity, and biodegradability make it suitable for many medical applications in cosmetic, biomedical materials, tissue engineering materials, and drug release fields [86–88]. The in vitro cytotoxicity of the collagen derived from tannery wastes has been studied [82]. Biocompatibility tests conducted in vitro demonstrated that the isolated collagen exhibited minimal cytotoxicity, with no observable cell death or growth inhibition. These findings may support the use of collagen derived from hide waste as a secure and economically viable alternative in diverse pharmaceutical and biomedical fields.

Gels for biomedical applications can be extracted from collagen sourced from the untanned waste of leather production, and collected at various stages of the production process, including limed pelt, delimed pelt, and fleshings [89]. The delimed pelt yielded the highest collagen content. Gels derived from these collagen extracts promote the adhesion and spreading of HEK293 cells [89]. This suggests that gel matrices based on collagen can serve as a substrate for cultivating biofilms in vitro. The biofilm infection in vitro wound model, comprising a 3D collagen sponge-like scaffold with HEK293 cells, could be used to replicate the intricate microenvironment of infected wounds for exploring therapeutic efficacy [90].

The leather of a donkey can be used for parchment or traditional footwear [91]. Ejiao, an age-old traditional Chinese medicine, is created through the process of stewing and concentrating from the hide of the donkey. For around 2000 years, Ejiao has been used as both a healthcare food and a constituent in traditional Chinese medicine [92]. Blood deficiency syndrome is a prevalent clinical syndrome in traditional Chinese medicine, denoting an insufficient blood pathological state. Ejiao demonstrates promise in alleviating blood deficiency in women without apparent adverse effects [93]. A study utilized Proton Nuclear Magnetic Resonance-based (1H NMR-based) metabolomics and correlation analysis to investigate metabolic profiles in anemic rats treated with Ejiao [94]. Varied serum biomarkers were identified, such as arginine and choline. The results indicated Ejiao's multi-target therapeutic effects on lipid metabolism, energy metabolism, gut microflora, and amino acid metabolism, shedding light on its hematinic mechanism in traditional Chinese medicine.

In the search for new antibiotics, prompted by the increasing prevalence of bacteria resistant to current treatments, fish skin mucus emerges as a promising source of diverse bioactive molecules, such as peptides and proteins, with antimicrobial properties [95]. Fish skin contains gelatin and collagen, which, when hydrolyzed, produce bioactive peptides with various beneficial properties [63]. Utilizing catheters in newborns poses potential risks, with possible infections and medication leaks. In treating a 36-week-old baby with a weight of 1600 g and a height of 40 cm, tilapia fish skin is used to help with wound healing and reduce complications [96]. Fish skin grafts offer a cost-effective alternative to allografts and xenografts. With the potential to treat various wounds, including diabetic foot ulcers and venous leg ulcers, fish skin grafts demonstrate promising material and biological properties [97].

4.2. Sensor

Conductive leather has garnered significant interest in capturing postures and monitoring human activities. Conductive leather has been developed through the in-situ polymerization of pyrrole, which can be used for making gloves capable of operating touch-screen devices [98,99]. A self-colored and conductive leather was produced by employing in-situ polymerization of aniline, utilizing ammonium persulfate as an oxidant and hydrochloric acid as a dopant [100].

A leather-based breath sensor utilizing ionic conductivity changes caused by humidity during breathing was developed [101]. By utilizing leather as a flexible base due to its biocompatibility, hydroscopicity, porosity, and ionic conductivity, and combining it with silver nanowires, it enables the easy non-invasive acquisition of human respiration rate, respiration depth, and respiration pattern. The application of pressure sensors spans various fields, including voice recognition, human motion detection, and artificial electronic skin (E-skin). The flexibility and stretchability of substrate materials significantly impact sensor performance. Innovative flexible pressure sensors have been developed using a pair of flexible and biocompatible collagen films, regenerated from leather waste [102]. A strain sensor utilizing a leather substrate was crafted by filtering an aqueous dispersion of conductive nanomaterials, which exhibited a distinctive microstructure, preserving the leather's qualities such as air permeability, mechanical properties, and biocompatibility [103]. Increasingly, electronic devices are being merged with biodegradable materials such as animal skin to lessen the environmental impact of e-waste. Animal-based porous substrates, known for their excellent compatibility with skin, are widely used in on-

Table 2
Several good properties of animal-based leather.

Property	Function	Reference
Breathability	It is a breathable material, allowing air to pass through and providing comfort in various climates.	[79]
Durability	It possesses inherent strength and durability due to its fibrous structure and natural composition	[68,77]
Flexibility	It is flexible and molds to the shape of the wearer or the item it covers.	[68,80,81]
Biocompatibility	It has remarkable biocompatibility, with its affinity for the human body, reducing the chances of adverse reactions.	[82]
Biodegradability	It is biodegradable and can naturally decompose over time, reducing environmental impact compared to synthetic materials.	[81,83]
Natural feel and aesthetics	Animal leather exudes innate elegance with its organic texture, evoking a sense of luxury and connection with nature.	[84]
Longevity	Animal-based leather endures through time, maintaining its quality and beauty.	[85]

skin electronics [81]. A leather-based E-skin was developed by incorporating various functional materials, including carbon nanotubes and silver nanowires, to impart sensing capabilities to the leather [104]. Fig. 3 depicts an application of animal-based leather on E-skin.

4.3. Medical implants

Sutures are natural or synthetic biomaterials, extensively employed for wound closure, ligating blood vessels, and bringing tissues together. For centuries, silk cocoons from Bombyx mori silkworms have been a valuable commodity utilized in textiles and medical sutures [105]. Silk leather is water-based, treated with mild chemicals, and manufactured at room temperature, reducing waste quantity and toxicity [106]. Silver/sercin coatings on absorbable sutures were developed on absorbable sutures to impart antibacterial and wound healing properties [107]. Absorbable sutures, developed with collagen from tannery waste and impregnated with a plasticizer, demonstrated biodegradability and high tensile strength [83].

Hybrid scaffolds were synthesized by utilizing type-I collagen extracted from leather industry wastes, Titanium dioxide (TiO2) nanoparticles, and cross-linked with oxidized sago starch from the agro-industry [108]. A nanobiocomposite was developed using collagen extracted from solid biowaste, specifically chrome-containing leather waste [109]. This material incorporates aluminum oxide nanoparticles for reinforcement, offering a straightforward and environmentally friendly approach to recycling and recovering biowaste. The collagen-based nanobiocomposites hold promise for applications such as hip prostheses and other biomedical uses.

Collagen from marine eel fish skin waste was characterized and tested for drug delivery potential. Type I collagen, with denaturation temperatures, demonstrated gel and film-forming capabilities for drug delivery [110]. Gelatin extracted from Pangasius skin and swim bladder waste could be utilized for hard capsule shells [111]. A technology for extracting chromium from leather industry wastes, enabling the use of chromium-free protein as a slow-release drug carrier for treating burns, ulcers, and infected wounds [112]. These biocompatible films show significant potential as drug carriers.

These features suggest the possibility of repurposing solid leather waste for enhanced medical products, such as sutures, scaffolds, and drug carriers, in biomedical applications. The medical applications discussed in this paper are summarized in Fig. 4.

4.4. Advantages and disadvantages

The biocompatibility, durability, and breathability are known as advantages of animal-based leather compared with synthetic leather. In addition to these advantages, collagen extracted from waste in the leather production process for medical applications can significantly contribute to environmental and economic benefits. It was estimated that raw hide trimmings, constituting 5 % of raw material in leather processing, contain at least 20 % collagen. From one metric ton of wet salted hides, around 200 kg of end product was obtained, with 250 kg chromium-containing solid waste, 350 kg chromium-free waste, and 100 kg lost in wastewater [113]. The waste is often dumped in developing or underdeveloped countries, but some nations utilize trimmings

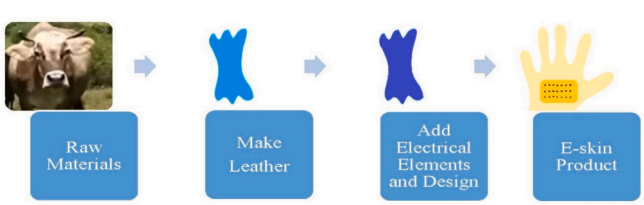

Fig. 3. Animal skin for producing E-skin.

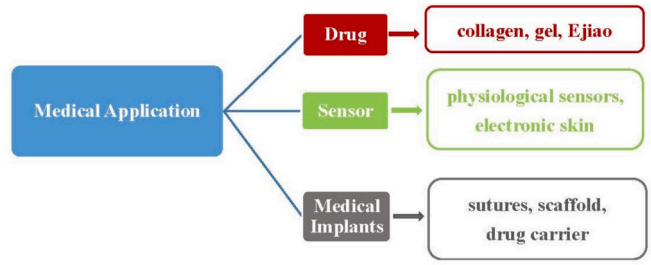

Fig. 4. The medical applications of animal-based leather or its waste reviewed in this paper.

for the production of glue, collagen peptides, gelatin, feed, and fertilizers. Therefore, effectively utilizing the waste from leather production to extract components for medical applications is a beneficial approach for waste reduction and economic considerations.

The collagen extracted from the leather production process for some medical applications can be obtained from the raw trimmings of the production waste [113] (Fig. 5).

Animal-based leather possesses several unique properties that contribute to its suitability for medical applications compared to synthetic or plant-based alternatives including biocompatibility, biodegradability, and durability. While synthetic and plant-based alternatives may offer advantages such as cost-effectiveness, the distinctive combination of properties offered by animal-based leather makes it the preferred choice for certain medical applications where biocompatibility, durability, and comfort are paramount.

While there are advantages to using leather in medical applications, there may also be disadvantages, such as cost, risk of contamination, and potential allergic reactions. The cost of animal-based leather continues to rise [114]. In addition, the animal-based leather may carry animal skin microbes, potentially causing allergic reactions in humans. Some microbes may withstand extreme conditions in leather manufacture and pose risks to humans. The link between animal leather and the human microbiome has been investigated [115].

Additionally, animal-based leather products may contain heavy metals such as arsenic, cadmium, chromium, copper, lead, and zinc as well as synthetic leather. A comparison of heavy metal levels in natural leathers, artificial leathers, and textiles has been conducted in the context of quality and safety of use [116]. Reducing the use of chemicals containing harmful metals in the leather manufacturing processes is suggested.

Overall, while leather has advantages for medical applications, careful consideration of its potential disadvantages is essential to ensure safety and effectiveness in healthcare settings.

5. Discussion

The global leather industry faces challenges such as limited raw materials, ethical concerns, competition from substitutes, and environmental regulations. While leather production has traditionally been viewed as an industry that pollutes the environment by employing harmful chemicals, recent years have witnessed significant

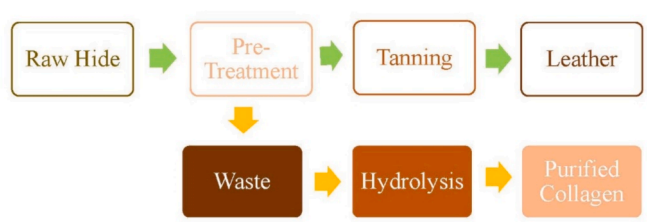

Fig. 5. The collagen extracted from the raw trimmings.

environmental advancements within leather-making companies [117]. Certain footwear and apparel companies have promoted vegetable-tanned leather as a product with a more environmentally friendly profile when compared to chromium-tanned leather [118]. In addition, nanotechnology offers potential solutions, enhancing product innovation and addressing environmental issues in various manufacturing stages [1].

Industries related to leather production generate plenty of proteinase biomass. Utilizing biomass for the production of bioenergy and biomaterials is viewed as a sustainable alternative to the depletion of fossil fuel resources. Leather biomass, primarily composed of 30 %–35 % collagen protein, can be effectively employed in the creation of various high-value materials, such as in cosmetic formulation, medical and pharmaceutical industries, food, edible food packaging, bioenergy, and biosurfactant synthesis applications [9].

In addition to medical applications, leather has many other applications. Functional leathers, including conductive, flame retardant, self-cleaning, antibacterial, waterproof, anti-fouling, oil-proof, electromagnetic shielding, thermal camouflage, and X-ray shielding leathers, have been successfully developed through combinations with other functional materials [3]. The exploration of cost-effective anode materials for sodium-ion batteries using a pyrolysis method applied to cow leather was undertaken [119]. Collagen bio-waste material, extracted from raw trimming wastes, was subjected to high-temperature treatment to facilitate the synthesis of a multifunctional nanocarbon structure, contributing to a sustainable approach to developing battery electrodes [120]. Pyrolysis temperature significantly impacts sodium storage, revealing the potential for converting cow waste into high-performance anode materials.

Furthermore, to prevent animal slaughter, a practical approach utilizing tissue engineering knowledge was developed to engineer skin for the creation of commercially viable leather-like material [121]. Advancements in synthetic biology and materials science have given rise to engineered living materials (ELMs), incorporating living matter in self-regenerating matrices. ELMs mimic natural materials, showing potential applications in green energy, bioremediation, disease treatment, and smart materials [122]. The traditional leather industry raises concerns about environmental, ethical, and health impacts. The production of leather-like materials by cellular agriculture may offer hope for sustainable, ethical, and environmentally friendly leather substitutes, but further research and development are required [123]. Since animal-based leather has unique advantages compared with other materials, it is hoped that the new biotechnologies can be developed to replicate very similar alternatives.

6. Conclusion

The leather industry, deeply rooted in ancient traditions, has long relied on animal skins, a rich source of collagen protein polymer, to produce versatile materials essential for various applications. While the use of animal-based leather has faced criticism for environmental and ethical reasons, its inherent benefits remain substantial in comparison to synthetic and plant-based vegan alternatives. This paper reviews the type of leather, mainly focusing on animal-based leather. The numerous merits of animal-based leather, when compared to synthetic and plant-based alternatives, are highlighted. These advantages include biodegradability, biocompatibility, a natural feel and aesthetics, breathability, durability, and longevity.

Moreover, exploring the potential medical applications of animal-based leather or its waste reveals a promising avenue for innovation. From breath sensors to electronic-sensing skin and gel, the unique properties of animal-based leather open new possibilities in the medical field. Since animal-based leather is commonly acknowledged for its superior quality and premium price, it is often utilized in the creation of luxurious items like bags, apparel, shoes, and sports gear. This paper aims to highlight the potential of animal-based leather in the production of medical equipment, which may provide significant benefits to humanity. In essence, despite the ongoing debate and the emergence of alternative materials, animal-based leather remains a crucial and irreplaceable material with a rich history of utility.

Funding.

This research was funded by the Ministry of Science and Technology (111–2118-M−A49−001−MY2), Taiwan.

Declaration of Generative AI and AI-assisted technologies in the writing process.

During the preparation of this work the author used ChatGPT in order to edit the language. After using this tool/service, the author reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Hsiuying Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The author declare the following financial interests/personal relationships which may be considered as potential competing interests: [Hsiuying Wang reports was provided by National Yang Ming Chiao Tung University. Hsiuying Wang reports a relationship with National Yang Ming Chiao Tung University that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper].

Data availability

No data was used for the research described in the article.

References

[1] L. Muthukrishnan, Nanotechnology for cleaner leather production: a review, Environmental Chemistry Letters 19 (3) (2021) 2527–2549.
[2] The New American Bible. 1991: Nashville: Catholic Bible Press.
[3] Y.Y. Wang, et al., Advanced collagen nanofibers-based functional bio-composites for high-value utilization of leather: a review, Journal of Science-Advanced Materials and Devices 6 (2) (2021) 153–166.
[4] K. Sai Bhavya, et al., Leather processing, its effects on environment and alternatives of chrome tanning, International Journal of Advanced Research in Engineering and Technology (IJARET) 10 (6) (2019).
[5] Y. Yu, et al., Life cycle assessment for chrome tanning, chrome-free metal tanning, and metal-free tanning systems, ACS Sustainable Chemistry & Engineering 9 (19) (2021) 6720–6731.
[6] C.R. China, et al., Alternative tanning technologies and their suitability in curbing environmental pollution from the leather industry: a comprehensive review, Chemosphere 254 (2020).
[7] A. Teklay, et al., Preparation of value added composite sheet from solid waste leather-A prototype design, Scientific Research and Essays 13 (2) (2018) 11–13.
[8] F.G. Nogueira, et al., Recycling of solid waste rich in organic nitrogen from leather industry: mineral nutrition of rice plants, Journal of Hazardous Materials 186 (2–3) (2011) 1064–1069.
[9] M. Ayele, et al., A review on utilization routes of the leather industry biomass, Advances in Materials Science and Engineering (2021) 1–15.
[10] G. Zhao, C. Liu, Carbon emission intensity embodied in trade and its driving factors from the perspective of global value chain, Environmental Science and Pollution Research 27 (25) (2020) 32062–32075.
[11] M. Ramchandani, I. Coste-Maniere, Leather in the age of sustainability: a norm or merely a cherry on top? Leather and Footwear Sustainability: Manufacturing, Supply Chain, and Product Level Issues (2020) 11–22.
[12] H. Wang, Beneficial medicinal effects and material applications of rose, Heliyon 10 (1) (2024) e23530.
[13] H. Wang, A review of the effects of collagen treatment in clinical studies, Polymers 13 (22) (2021) 3868.
[14] H. Wang, The potential of collagen treatment for comorbid diseases, Polymers 15 (19) (2023) 3999.
[15] H. Wang, Biomaterials in medical applications, Polymers 15 (4) (2023) 847.

[16] Maina, P., M.A. Ollengo, and E.W. Nthiga, Trends in leather processing: A Review. 2019.
[17] B. Amde, Major factors affecting hide and skin production, quality and the tanning industry in Ethiopia, Advances in Biological Research 11 (3) (2017) 116–125.
[18] B. Haines, J. Barlow, The anatomy of leather, Journal of Materials Science 10 (1975) 525–538.
[19] A. Mammadova, M. Masiero, D. Pettenella, Embedded deforestation: the case study of the Brazilian-Italian bovine leather trade, Forests 11 (4) (2020) 472.
[20] M. Adem, Production of hide and skin in Ethiopia; marketing opportunities and constraints: a review paper, Cogent Food & Agriculture 5 (1) (2019) 1565078.
[21] S. Cloete, et al., Effect of age on leather and skin traits of slaughter ostriches, South African Journal of Animal Science 34 (2) (2004) 80–86.
[22] S.J. Van Schalkwyk, Factors affecting ostrich leather traits, Stellenbosch University, Stellenbosch, 2008.
[23] A. Nasr, Shedding light on camel leather, Journal of Camelid Science 13 (2020) 1–9.
[24] M.H. Ahmed, A. Musa, E.A. Hassan, Manufacture of shoe upper leathers from sudanese camel hides, Sch J EngTech 7 (2022) 146–152.
[25] S.A. Hekal, Histological study of the skin and leather characteristics in two types of Arabian camels (Camelus dromedarius), The Journal of American Science 10 (9) (2014) 41–48.
[26] M. Habib, A. Alhamzani, A.I. Alakhras, Leather fatliquoring agent from camel hump fat, Egyptian Journal of Chemistry 65 (131) (2022) 379–383.
[27] R. Duraisamy, S. Shamena, A.K. Berekete, A review of bio-tanning materials for processing of fish skin into leather, International Journal of Engineering Trends and Technology 39 (1) (2016) 10–20.
[28] M. Fadda, et al., Hydrophobic and water resistant fish leather: a fully sustainable combination of discarded biomass and by-products of the food industry, Green Chemistry (2024) 542–555.
[29] J.P. Alla, et al., Fish skin and exotic leathers, Journal of the American Leather Chemists Association 112 (2) (2017) 36–43.
[30] A. Zilberfarb, G. Cohen, E. Amir, Increasing Functionality of Fish Leather by Chemical Surface Modifications, Polymers 15 (19) (2023) 3904.
[31] P.N. Wainaina, B. Ongarora, P. Tanui, Manufacture of exotic leather and small leather goods from ovine stomach, Skin 30 (2022) 6.
[32] F. Ali, et al., Manufacturing of novelty leather from goat stomach, INTERNATIONAL JOURNAL OF CURRENT RESEARCH 11 (07) (2019) 5156–5160.
[33] T.B. Sudha, et al., Comfort, chemical, mechanical, and structural properties of natural and synthetic leathers used for apparel, Journal of Applied Polymer Science 114 (3) (2009) 1761–1767.
[34] M. Chen, et al., Analyses of structures for a synthetic leather made of polyurethane and microfiber, Journal of Applied Polymer Science 103 (2) (2007) 903–908.
[35] T. Sano, S. Suzuki, Basic forensic identification of artificial leather for hit-and-run cases, Forensic Science International 192 (1–3) (2009) e27–e32.
[36] A.M. Neiva, E.R. Pereira, Evaluation of the chemical composition of synthetic leather using spectroscopy techniques, Applied Spectroscopy 72 (6) (2018) 921–932.
[37] Y.J. Tian, et al., Research on the preparation and application of synthetic leather from coffee grounds for sustainable development, Sustainability 14 (21) (2022) 13971.
[38] X. Chen, et al., Sustainable supply chain management in the leather industry: a systematic literature review, International Journal of Logistics Research and Applications 26 (12) (2023) 1663–1703.
[39] W. Kim, A study on the subjective feeling affecting tactile satisfaction of leather in automobile: a structural equation modeling approach, International Journal of Industrial Ergonomics 84 (2021) 103167.
[40] S. Tewari, et al., Vegan leather: a sustainable reality or a marketing gimmick? Environmental Science and Pollution Research (2023) 1–15.
[41] Z. Sun, et al., A self-matting waterborne polyurethane coating for PVC artificial leather, Polymers 15 (1) (2022) 127.
[42] X. Zhu, et al., Current advances of Polyurethane/Graphene composites and its prospects in synthetic leather: a review, European Polymer Journal 161 (2021) 110837.
[43] S. Tian, Recent advances in functional polyurethane and its application in leather manufacture: a review, Polymers 12 (9) (2020) 1996.
[44] S. Lu, et al., Preparation of flame-retardant polyurethane and its applications in the leather industry, Polymers 13 (11) (2021) 1730.
[45] Kaur, S., et al. (2018) FAUX leather exploration: A creative approach. 4: p. 1747-1751.
[46] M. Xu, et al., Perspective on the disposal of PVC artificial leather via pyrolysis: thermodynamics, kinetics, synergistic effects and reactionmechanism, Fuel 327 (2022).
[47] Minh, N.T. and H.N. Ngan. Vegan leather: An eco-friendly material for sustainable fashion towards environmental awareness. in AIP Conference Proceedings. 2021. AIP Publishing.
[48] E. Cherry, Veganism as a cultural movement: a relational approach, Social Movement Studies 5 (2) (2006) 155–170.
[49] S. Duangsuwan, et al., Development of green leather alternative from natural rubber and pineapple leaf fiber, Sustainability 15 (21) (2023) 15400.
[50] H.J. Jung, H. Kim, K.W. Oh, Green leather for ethical consumers in China and Korea: facilitating ethical consumption with value–belief–attitude logic, Journal of Business Ethics 135 (2016) 483–502.
[51] Y.H. Choi, K.H. Lee, Ethical consumers' awareness of vegan materials: focused on fake fur and fake leather, Sustainability 13 (1) (2021) 436.
[52] M. Jones, et al., Leather-like material biofabrication using fungi, Nature Sustainability 4 (1) (2021) 9–16.
[53] A. Amobonye, et al., Fungal mycelium as leather alternative: a sustainable biogenic material for the fashion industry, Sustainable Materials and Technologies (2023).
[54] C. Wjunow, et al., Sustainable textiles from unconventional biomaterials—cactus based, Engineering Proceedings 37 (1) (2023) 58.
[55] G.Y. Kefale, Z.T. Kebede, A.A. Birlie, A systematic review on potential bio leather substitute for natural leather, Journal of Engineering (2023).
[56] R. da Silva Simão, et al., Production of mango leathers by cast-tape drying: product characteristics and sensory evaluation, Lwt 99 (2019) 445–452.
[57] S.P. Singaraj, et al., Investigations on structural, mechanical and thermal properties of banana fabrics for use in leather goods application, Journal of Natural Fibers 18 (11) (2021) 1618–1628.
[58] A.D. Covington, W.R. Wise, Current trends in leather science, Journal of Leather Science and Engineering 2 (2020) 1–9.
[59] Y. Zhang, et al., Effect of different ions in assisting protease to open the collagen fiber bundles in leather making, Journal of Cleaner Production 293 (2021).
[60] M.M. Hassan, et al., A review of the green chemistry approaches to leather tanning in imparting sustainable leather manufacturing, Green Chemistry 25 (19) (2023) 7441–7469.
[61] L. Falcao, M.E.M. Araújo, Vegetable tannins used in the manufacture of historic leathers, Molecules 23 (5) (2018) 1081.
[62] P. Auad, F. Spier, M. Gutterres, Vegetable tannin composition and its association with the leather tanning effect, Chemical Engineering Communications 207 (5) (2020) 722–732.
[63] R. Abuine, A.U. Rathnayake, H.-G. Byun, Biological activity of peptides purified from fish skin hydrolysates, Fisheries and Aquatic Sciences 22 (1) (2019) 1–14.
[64] S. Ganguly, Quantum Dots-Rubber Composites, in: Quantum Dots and Polymer Nanocomposites, CRC Press, 2022, pp. 189–205.
[65] S. Ganguly, P. Das, N.C. Das, Characterization tools and techniques of hydrogels, in: Hydrogels Based on Natural Polymers, Elsevier, 2020, pp. 481–517.
[66] Mazotto, A.M., Biopolymers in the leather industry, in Biopolymers: Recent Updates, Challenges and Opportunities. 2022, Springer. p. 245-270.
[67] Y. Mahmud, et al., Enhancement of antimicrobial properties of shoe lining leather using chitosan in leather finishing, Advances in Materials Research 9 (3) (2020) 233–250.
[68] S.J.R. Kelly, et al., Effect of collagen packing and moisture content on leather stiffness, Journal of the Mechanical Behavior of Biomedical Materials 90 (2019) 1–10.
[69] Y. Kumazawa, et al., A rapid and simple LC-MS method using collagen markerpeptides for identification of the animal source of leather, Journal of Agricultural and Food Chemistry 64 (30) (2016) 6051–6057.
[70] H.C. Wells, et al., Collagen fibril diameter and leather strength, Journal of Agricultural and Food Chemistry 61 (47) (2013) 11524–11531.
[71] S. Selvaraju, S. Ramalingam, J.R. Rao, Preparation and application of biodegradable nanocomposite for cleaner leather processing, Journal of Cleaner Production 158 (2017) 225–232.
[72] C.E.M. Braz, et al., Potential of near-infrared spectroscopy for quality evaluation of cattle leather, Spectrochimica Acta Part a-Molecular and Biomolecular Spectroscopy 202 (2018) 182–186.
[73] D.A. Van der Merwe, et al., Sheepskin leather quality characteristics of South African breeds, Small Ruminant Research 199 (2021) 106365.
[74] F. Negussie, et al., Effects of different feeding regimes on leather quality of finished blackhead ogaden sheep, Science, Technology and Arts Research Journal 4 (2) (2015) 222–227.
[75] 75.Catauro, M., et al. Use of Spectroscopic, Mechanical, and Microbiological Analysis to Study a Natural Sheep Leather. in Macromolecular Symposia. 2023. Wiley Online Library.
[76] H. Memon, et al., Study on effect of leather rigidity and thickness on drapability of sheep garment leather, Materials 14 (16) (2021) 4553.
[77] M. Meyer, et al., Comparison of the technical performance of leather, artificial leather, and trendy alternatives, Coatings 11 (2) (2021) 226.
[78] E. Saad, Study of different methods of cleaning and care of leather clothing, International Design Journal 10 (1) (2020) 339–347.
[79] N.-T. Nguyen, T.-H. Vu, V.-H. Bui, Antibacterial and antifungal fabrication of natural lining leather using bio-synthesized silver nanoparticles from piper betle L, Leaf Extract. Polymers 15 (12) (2023) 2634.
[80] P. Kesarwani, S. Jahan, K. Kesarwani, A review on leatherprocessing, International Journal of Applied Research 1 (9) (2015) 977–982.
[81] Y. Zhang, et al., A new class of electronic devices based on flexible porous substrates, Advanced Science 9 (7) (2022) 2105084.
[82] B.R. Rajan durai, V. Sankar, J. Koilpillai, Isolation and characterization of collagen from tannery waste for biomedical applications, BioNanoScience (2023) 1–16.
[83] S. Rethinam, et al., Preparation of absorbable surgical suture: novel approach in biomedical application, Journal of Drug Delivery Science and Technology 47 (2018) 454–460.
[84] S. Watanabe, S. Tominaga, T. Horiuchi, The difference in impression between genuine and artificial leather: quantifying the feeling of authenticity, Electronic Imaging 33 (2020) 1–11.
[85] M. Ramchandani, I. Coste-Maniere, To Fur or not to fur: Sustainable production and consumption within animal-based luxury and fashion products, Sustainable Fashion and Consumption, Textiles and Clothing Sustainability, 2017, pp. 41–60.

[86] Y. Han, J. Hu, G. Sun, Recent advances in skin collagen: functionality and non-medical applications, Journal of Leather Science and Engineering 3 (2021) 1–12.
[87] B. Yu, et al., An elastic second skin, Nature Materials 15 (8) (2016) 911–918.
[88] M.Y. Li, et al., Electrospun protein fibers as matrices for tissue engineering, Biomaterials 26 (30) (2005) 5999–6008.
[89] L. Maistrenko, et al., Collagen obtained from leather production waste provides suitable gels for biomedical applications, Polymers 14 (21) (2022) 4749.
[90] O.V. Moshynets, et al., Therapeutic potential of an azithromycin-colistin combination against XDR in a 3D collagen-based in vitro wound model of a biofilm infection, Antibiotics-Basel 12 (2) (2023) 293.
[91] W. Kugler, H.-P. Grunenfelder, E. Broxham, Donkey breeds in Europe, St. Gallen, Switzerland, 2008.
[92] J. Du, et al., Analysis of Heparan sulfate/heparin from Colla corii asini by liquid chromatography-electrospray ion trap mass spectrometry, Glycoconjugate Journal 36 (2019) 211–218.
[93] L. Zhang, et al., Efficacy and safety of Ejiao (Asini Corii Colla) in women with blood deficient symptoms: a randomized, double-blind, and placebo-controlled clinical trial, Front Pharmacol 12 (2021) 718154.
[94] J.-S. Tian, et al., The hematinic effect of Colla corii asini (Ejiao) using 1 H-NMR metabolomics coupled with correlation analysis in APH-induced anemic rats, RSC Advances 7 (15) (2017) 8952–8962.
[95] R. Díaz-Puertas, et al., Fish skin mucus extracts: an underexplored source of antimicrobial agents, Marine Drugs 21 (6) (2023) 350.
[96] N. Faraji, et al., Treatment of severe extravasation injury in a newborn by using tilapia fish skin: a case report, International Journal of Surgery Case Reports 91 (2022) 106759.
[97] G. Fiakos, Z. Kuang, E. Lo, Improved skin regeneration with acellular fish skin grafts, Engineered Regeneration 1 (2020) 95–101.
[98] J.D. Wegene, P. Thanikaivelan, Conducting leathers for smart product applications, Industrial & Engineering Chemistry Research 53 (47) (2014) 18209–18215.
[99] Z. Bai, et al., Leather for flexible multifunctional bio-based materials: a review, Journal of Leather Science and Engineering 4 (1) (2022) 16.
[100] W.J. Demisie, et al., Concurrent genesis of color and electrical conductivity in leathers through in-situ polymerization of aniline for smart product applications, Polymers for Advanced Technologies 26 (5) (2015) 521–527.
[101] R. Xie, et al., Wearable leather-based electronics for respiration monitoring, ACS Appl BioMater 2 (4) (2019) 1427–1431.
[102] J. Lei, et al., Regenerating leather waste for flexible pressure sensing applications, Journal of Leather Science and Engineering 1 (1) (2019) 1–9.
[103] R.J. Xie, et al., Leather-based strain sensor with hierarchical structure for motion monitoring, Advanced Materials Technologies 4 (10) (2019) 1900442.
[104] B.H. Zou, et al., Repurposed leather with sensing capabilities for multifunctional electronic skin, Advanced Science 6 (3) (2019) 1801283.
[105] F.G. Omenetto, D.L. Kaplan, New opportunities for an ancient material, Science 329 (5991) (2010) 528–531.
[106] L. Mogas-Soldevila, et al., Additively manufactured leather-like silk protein materials, Materials & Design 203 (2021) 109631.
[107] A.L. Gallo, M. Pollini, F. Paladini, A combined approach for the development of novel sutures with antibacterial and regenerative properties: the role of silver and silk sericin functionalization, Journal of Materials Science: Materials in Medicine 29 (2018) 1–13.
[108] C. Kalirajan, et al., A facile approach to fabricate dual purpose hybrid materials for tissue engineering and water remediation, Scientific Reports 9 (1) (2019) 1040.
[109] A. Mandal, M. Katheem Farhan, T.P. Sastry, Effect of reinforced Al 2 O 3 nanoparticles on collagen nanobiocomposite from chrome-containing leather waste for biomedical applications, Clean Technologies and Environmental Policy 18 (2016) 765–773.
[110] A. Veeruraj, et al., Isolation and characterization of drug delivering potential of type-I collagen from eel fish Evenchelys macrura, Journal of Materials Science: Materials in Medicine 23 (2012) 1729–1738.
[111] M. Nurilmala, et al., Pangasius fish skin and swim bladder as gelatin sources for hard capsule material, International Journal of Biomaterials (2021).
[112] C.S. Gil, et al., Recycled collagen films as biomaterials for controlled drug delivery, New Journal of Chemistry 40 (10) (2016) 8502–8510.
[113] D. Masilamani, et al., Extraction of collagen from raw trimming wastes of tannery: a waste to wealth approach, Journal of Cleaner Production 113 (2016) 338–344.
[114] H.T. Nguyen, et al., Silane-modified kombucha-derived cellulose/polyurethane/polylactic acid biocomposites for prospective application as leather alternative, Sustainable Materials and Technologies 36 (2023) e00611.
[115] R.O. Oruko, J.O. Odiyo, J.N. Edokpayi, The role of leather microbes in human health, Role of Microbes in Human Health and Diseases (2019) 243–371.
[116] E. Bielak, E. Marcinkowska, Heavy metals in leathers, artificial leathers, and textiles in the context of quality and safety of use, Scientific Reports 12 (1) (2022) 5061.
[117] R. Laurenti, et al., Measuring the environmental footprint of leather processing technologies, Journal of Industrial Ecology 21 (5) (2017) 1180–1187.
[118] M.A. Achabou, S. Dekhili, Luxury and sustainable development: is there a match? Journal of Business Research 66 (10) (2013) 1896–1903.
[119] X. Chen, et al., Cow leather-derived N/O codoped hard carbon as a high-performance anode material for sodium-ion batteries, Materials Today Communications 37 (2023) 107361.
[120] M. Ashokkumar, et al., Transforming collagen wastes into doped nanocarbons for sustainable energy applications, Green Chemistry 14 (6) (2012) 1689–1695.
[121] T.-H. Chuang, Non-medical applications of tissue engineering: biofabrication of a leather-like material, Materials Today Sustainability 5 (2019) 100018.
[122] B. An, et al., Engineered living materials for sustainability, Chemical Reviews 123 (5) (2022) 2349–2419.
[123] D. Wilson, et al., Leather-like materials by cellular agriculture, Textile Progress 55 (1) (2023) 1–45.